Yasmine Mustafa, PhD

Yasmine Mustafa Hussain Ahmed Attia

6th of October, Giza | | | |

Summary

Tenure-track Assistant Professor at the American University in Cairo with a Ph.D. in Computer Science from Missouri University of Science and Technology.

Education

Missouri University of Science and Technology, PhD in Computer Science — Rolla, MO — Jan 2023 – May 2026

Dissertation focus: Anatomically-Informed Interpretable Multimodal Learning for Alzheimer's Disease Diagnosis and Progression

- Developed novel transformer architectures for trustworthy multimodal machine learning.
- Proposed explainable AI methods for interpretable medical image analysis.
- Designed scalable deep learning pipelines using Vision Transformers, foundation models, and large language models.

German University in Cairo, M.Sc. in Digital Media Engineering and Technology — Cairo, Egypt — Mar 2021 – Jan 2022

Thesis: Brain–Computer Interface for Augmented Reality | Grade: A+

- Collected and processed EEG data for real-time brain–computer interface applications.
- Developed ensemble models for decoding SSVEP signals and predicting user intent from neural data.
- Analyzed neural responses to visual stimuli for attention modeling and signal characterization.

German University in Cairo, B.Sc. in Digital Media Engineering and Technology — Cairo, Egypt — Sept 2015 – Jul 2020

GPA: A+ | Semester Abroad: GUC Berlin, Germany (Sept 2019 – Jan 2020), GPA: A+
Bachelor Thesis Abroad: Fraunhofer IGD, Rostock, Germany (Mar 2019 – Aug 2019)
Thesis: Design and Development of a Telemetric Monitoring System for Joint Mobility of Knee or Hip using Machine Learning | Grade: A+

Academic Appointments

Assistant Professor (Tenure-Track), The American University in Cairo — New Cairo, Egypt — Aug 2026 – Present
Department of Mathematics and Actuarial Science

Research Funding & Awards

Best Paper Award — PAKDD 2025 Workshop on Pattern Mining and Machine Learning for Bioinformatics — 2025

- Awarded for "*Unlocking Neural Transparency: Jacobian Maps for Explainable AI in Alzheimer's Detection*".

John W. Claypool Fund for Medical Research Fellowship — Missouri University of Science and Technology — 2024

- Competitive fellowship supporting research in applied artificial intelligence.

First Place — 3 Minute Thesis (3MT) — Missouri University of Science and Technology — 2024

- First place in campus-wide competition; advanced to regional MAGS 3MT.

Research Experience

Multimodal Machine Learning for Behavioral and Health Data — Missouri S&T & Saint Louis University Jan 2026 – Present

- Developed multimodal AI models integrating biological and behavioral data for predictive health analytics under missing-modality settings.
- Co-authored two manuscripts currently under preparation through interdisciplinary collaboration with Missouri S&T and Saint Louis University.

Research Collaborator — Tellegacy Sep 2025 – Present

- Conducted interdisciplinary AI research at the intersection of machine learning, healthcare, and behavioral science.
- Produced **three research publications**, including **one peer-reviewed publication** and **two manuscripts currently under review**.

Teaching Experience

Teaching Assistant, Washington State University — Washington, USA Feb 2026 – May 2026

- Supported instruction in cybersecurity and web development courses through student mentoring, technical guidance, grading, and classroom support.
- Assisted students with programming, software implementation, debugging, and applied computing concepts.
- Proctored exams, evaluated assignments, and supported course administration.

Teaching Assistant, Missouri University of Science and Technology — Missouri, USA Feb 2026 – May 2026

- Assessed undergraduate research reports and technical assignments using structured rubrics.
- Mentored students in research methodology, experimental design, and technical communication.
- Contributed to curriculum support and assessment aligned with course learning outcomes.

Teaching Assistant, German University in Cairo — Cairo, Egypt Mar 2021 – Jan 2022

- Assisted in teaching Data Structures and Algorithms and Introduction to Computer Programming (Java).
- Guided undergraduate students through programming assignments, debugging exercises, and algorithmic problem solving.
- Supported laboratory instruction and evaluated coursework and programming projects.
- Mentored students in foundational programming concepts and software development practices.

Industry Experience

Full-Time Software Engineer, Everteam Intalio — Cairo, Egypt Feb 2021 – Mar 2021

- Developed enterprise .NET software while improving system reliability and application performance.

Part-Time Full Stack Developer, Lirten Solutions — Cairo, Egypt Aug 2020 – Jan 2021

- Built full-stack web applications using React, Node.js, MongoDB, and PostgreSQL.

Internships

Deep Learning Intern, ARAtronics Lab (Robotics) — Cairo, Egypt Jul 2018

- Developed a deep learning model to control a prosthetic arm using Myo armband signals (MATLAB, Neural Networks Toolbox).

Web Developer Intern, German University in Cairo — Cairo, Egypt Jun 2018

- Developed a website for the Human++ research cluster using WordPress, PHP, and JavaScript frameworks.

Speech Recognition Intern, German University in Cairo — Cairo, Egypt Jul 2017

- Collected and processed audio data for speech recognition experiments.

Teaching Interests

Artificial Intelligence, Machine Learning, Data Science, Business Analytics, Deep Learning, Computer Vision, Natural Language Processing, Generative AI, Explainable AI, Programming, Data Structures and Algorithms, Software Engineering, Database Systems, Responsible AI.

Research Areas

Machine Learning, Deep Learning, Explainable AI, Computer Vision, Multimodal Learning, Foundation Models, Large Language Models, Generative AI, Medical AI, Data Mining, Predictive Analytics, Trustworthy AI.

Professional Service

- Program Committee Member, Association for the Advancement of Artificial Intelligence Conference (AAAI 2027).
- Invited Reviewer, International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) — reviewed 10+ submissions across multiple years; recognized on official reviewer list.
- Reviewer, Association for the Advancement of Artificial Intelligence (AAAI) — reviewed 5+ submissions.
- Reviewed manuscripts in AI, medical imaging, and machine learning with emphasis on experimental rigor and reproducibility.

Publications

1. **Yasmine Mustafa**, Haojie Zhu, Xiaowei Yu, and Seung-Jong Park. *Anatomical Attention Transformer for Region-Wise Fusion of 3D Neuroimaging and Clinical Text in Alzheimer's Disease*, IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 2026.
2. **Yasmine Mustafa**, Xiaowei Yu, and Seung-Jong Park. *Anatomically Informed and Compressed 3D Vision Transformer for Dementia Detection*. Under review, BMC Bioinformatics, 2026.
3. **Yasmine Mustafa**, Haojie Zhu, Xiaowei Yu, and Seung-Jong Park. *Forecast-Guided Multimodal Fusion via Latent Transition Modeling for Alzheimer's Disease Progression*. Under review, ACM International Conference on Multimedia (ACM Multimedia), 2026.
4. J. Holloway, D. Gans, N. Akram, D. O'Brien, A. Vincent, L. Johnson, **Yasmine Mustafa**, O. Sayeed, and Z. S. Tan. *Lessons Learned from an Intergenerational Health Promotion Program Focused on Virtual Community Engagement*. Discover Public Health, vol. 23, article 844, 2026.
5. **Yasmine Mustafa**, M. Elmahallawy, and T. Luo. *Unlocking Neural Transparency: Jacobian Maps for Explainable AI in Alzheimer's Detection*. Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2025.
6. **Yasmine Mustafa**, M. Elmahallawy, and T. Luo. *Efficient Brain Imaging Analysis for Alzheimer's and Dementia Detection Using Convolution-Derivative Operations*. IEEE International Conference on Big Data (BigData), 2024, pp. 6420–6429.
7. **Yasmine Mustafa**, and T. Luo. *Unmasking Dementia Detection by Masking Input Gradients: A JSM Approach to Model Interpretability and Precision*. Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2024, pp. 75–90.
8. **Yasmine Mustafa**, and T. Luo. *Diagnosing Alzheimer's Disease using Early-Late Multimodal Data Fusion with Jacobian Maps*. IEEE International Conference on E-health Networking, Applications and Services (Healthcom), 2023.
9. **Yasmine Mustafa**, M. Elmahallawy, T. Luo, and S. Eldawlatly. *A Brain-Computer Interface Augmented Reality Framework with Auto-Adaptive SSVEP Recognition*. Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRAINE), 2023.
10. **Yasmine Mustafa**, and S. Eldawlatly. *An Ensemble Classification Approach for Recognizing Steady-State Visually Evoked Potentials Frequencies*. IEEE International Conference on Computer Engineering and Systems (ICCES), 2021, pp. 1–6.